Lithium-Ion Chemistry Comparison

Description	Lithium Iron Phosphate (LFP)	Lithium Nickel Manganese Cobalt Oxide (NMC)	Lithium Manganese Oxide (LMO) (may contain Cobalt)	Lithium Nickel Cobalt Aluminium Oxide (NCA)	Lithium Cobalt Oxide (LCO)
Danger of thermal runaway and fire	No	Yes	Yes	Yes	Yes
Toxic elements	No	Yes	Yes	Yes	Yes
Landfill safe	Yes	No	No	No	No
Involves abusive mining practices	No	Yes	Yes	Yes	Yes
Retained capacity/cycle life	80%/10,000 cycles	60%/10,000 cycles	60%/5,000 cycles	70%/10,000 cycles	80%/8,000 cycles
C-rate to maintain warranty	C/2	C/5	C/2	C/2	C/2
Ventilation required	No	Yes	Yes	Yes	Yes
Cooling equipment required	No	Yes	Yes	Yes	Yes
Safety monitoring equipment required	No	Yes	Yes	Yes	Yes
Able to withstand high temperature environments	Yes (up to 140°)	No	No	No	No

(Source: https://simpliphipower.com)

damage from shock and vibration during transport and use. It should also minimise damage during vehicle collisions. Multiple insulation barriers should be considered where high voltages are present. A flame-retardant compound should be used to suppress or delay the production of flames, or prevent the spread of fire.

The simplest form of protection in an electric vehicle (EV) against thermal runaway is to place protective material between the battery pack and chassis. The next form of protection against thermal runaway is to place protective material between battery modules. A thermal conducting film, thermal conducting plates, fans and liquid cooling systems are used to cool down the battery or to dissipate heat from a hot spot.

Battery system design can reduce thermal runaway

Cell manufacturers have lately modified lithium-ion cell chemistries of the electrolyte, anode, cathode, separator and case to reduce the incidence and intensity of thermal runaway. They all are now designed to maximise safety and temperature range while attempting to maintain or increase cell watt-hour capacity. However, these changes come with some trade-offs such as lower cell capacity or higher cost.

Dr Daniele Suzzi, lead engineer - HV battery and EE thermal, AVL List GmbH, says, "Safety concern is one of the major issues during development and homologation of EVs with lithium-ion battery systems. Therefore thermal propagation must be effectively considered in battery safety design, with both support of numerical simulation and real-life tests."

Thermal fuses are good solutions that can be reflow-soldered at 260°C and still open at 210°C. These include high operating current (up to 100A), high rated voltage (60V DC), low resistance (120µ) and very high breaking capacity.

AllCell's PCC material is designed to prevent thermal runaway from propagating throughout the battery pack. When thermal runaway occurs in a single cell, wax in the PCC rapidly absorbs a large amount of heat, while the graphite spreads the remaining heat evenly throughout the entire pack.

Key to an efficient lithium-ion battery is to conduct cell selection and battery system design according to the target application to facilitate safety, performance and cost-efficiency. Mechanical and thermal stability of the battery is ensured by appropriate monitoring mechanisms of battery cells and the battery pack.

Temperature sensors on every block connected to a battery monitoring system allow for early detection of thermal runaway. If a permanently-installed monitor takes daily temperature differential readings and floats current readings, the battery can never go to thermal runaway without the monitor first detecting and alarming the condition.

Fans are available in various sizes, some with integrated pulse width modulation (PWM), speed measurement, speedometer signal or automatic restart for better results.

Battery management system design for safety

A battery management system manages the battery within acceptable performance and safe operating conditions. It requires precise measurement of cell voltage, charge and discharge currents, and temperature. As all of these could cause a rise in cell temperature, cells are also equipped with a polyswitch that prevents excessively high currents out of the cell. Cells are equipped with a burst valve that opens when the pressure gets too high, or the battery is exposed to heat externally, to reduce the temperature and prevent thermal runaway.

It is important to consider the cell's shape when specifying cell protection, as different cells have different insulation needs. There are three main shapes for cells, namely, cylindrical, prismatic and pouch. There are two options for thermal protection in automotive, namely, active or passive management.